

Unit 2.1

Ex 2.1 p82

1(a) $P(15, 250)$

t	Q	mpq
5	(5, 694)	-44.4
10	(10, 444)	-38.8
20	(20, 111)	-27.8
25	(25, 28)	-22.2
30	(30, 0)	-16.6

(b) Using $t=10$ and $t=20$ to approximate $t=15$:

$$\frac{-38.8 + (-27.8)}{2} = -33.3$$

8. $s = 25 \sin \pi t + 3 \cos \pi t$

(a) (i) $\frac{s(2) - s(1)}{2-1} = 6 \text{ cm/s}$

(ii) $\frac{s(1.1) - s(1)}{0.1} = -4.71 \text{ cm/s}$

(iii) $\frac{s(1.01) - s(1)}{0.01} = -6.13 \text{ cm/s}$

(iv) $\frac{s(1.001) - s(1)}{0.001} = -6.27 \text{ cm/s}$

(b) Instantaneous velocity if $t=1$ is approx. -6.3 cm/s .

(students)

Unit 2.2

Ex 2.2 p92

1. $\lim_{x \rightarrow 2} f(x) = 5$ — As x approaches 2, $f(x)$ approaches 5.

2. $\lim_{x \rightarrow 2^-} f(x) = 1$ — As x approaches 2 from the left, $f(x)$ approaches 1.

$\lim_{x \rightarrow 2^+} f(x) = 5$ — As x approaches 2 from the right, $f(x)$ approaches 5.

No limit exists because left- and right limits are different.

5. (a) $\lim_{x \rightarrow 1} f(x) = 2$

(b) $\lim_{x \rightarrow 3^-} f(x) = 1$

(c) $\lim_{x \rightarrow 3^+} f(x) = 4$

(d) $\lim_{x \rightarrow 3} f(x) \text{ DNE}$

(e) $f(3) = 3$

33. $\lim_{x \rightarrow 1} \frac{2-x}{(x-1)^2} = \infty$

34. $\lim_{x \rightarrow 3^-} \frac{\sqrt{x}}{(x-3)^5} = -\infty$

37. $\lim_{x \rightarrow \pi/2^+} \frac{1}{x} \sec x = -\infty$
 $\searrow + \quad \searrow -\infty$

38. $\lim_{x \rightarrow \pi^-} \cot x = -\infty$

Unit 2.3

Ex 2.3 p102

$$\begin{aligned} 2. (a) \lim_{x \rightarrow 2} [f(x) + g(x)] \\ = \lim_{x \rightarrow 2} f(x) + \lim_{x \rightarrow 2} g(x) \\ = -1 + 2 = 1 \end{aligned}$$

$$(b) \lim_{x \rightarrow 0} g(x) \text{ DNE so } \lim_{x \rightarrow 0} [f(x) - g(x)] \text{ DNE.}$$

$$\begin{aligned} (c) \lim_{x \rightarrow -1} [f(x) \cdot g(x)] \\ = \lim_{x \rightarrow -1} f(x) \cdot \lim_{x \rightarrow -1} g(x) \\ = 1 \cdot 2 = 2 \end{aligned}$$

$$(d) \lim_{x \rightarrow 3} \left[\frac{f(x)}{g(x)} \right] \text{ DNE since } \lim_{x \rightarrow 3} g(x) = 0$$

$$\begin{aligned} (e) \lim_{x \rightarrow 2} [x^2 f(x)] \\ = \lim_{x \rightarrow 2} x^2 \cdot \lim_{x \rightarrow 2} f(x) \\ = 2^2 \cdot (-1) = -4 \end{aligned}$$

$$(f) f(-1) + \lim_{x \rightarrow -1} g(x) \text{ is undefined since } f(-1) \text{ undefined.}$$

$$13. \lim_{x \rightarrow 2} \frac{x^2 - x + 6}{x - 2} \text{ DNE}$$

Numerator $\frac{x}{2} \rightarrow 8$ Denominator $\frac{x}{2} \rightarrow 0$

$$\begin{aligned} 40. -1 \leq \sin\left(\frac{\pi}{x}\right) \leq 1 \\ \Rightarrow e^{-1} \leq e^{\sin(\pi/x)} \leq e \\ \Rightarrow \frac{\sqrt{x}}{e} \leq \sqrt{x} e^{\sin(\pi/x)} \leq e\sqrt{x} \end{aligned}$$

$$\begin{aligned} \text{Since } \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{e} = \lim_{x \rightarrow 0^+} e\sqrt{x} = 0 \\ \text{by the squeeze theorem} \\ \lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin(\pi/x)} = 0 \end{aligned}$$

$$41. |x-3| = \begin{cases} x-3 & \text{if } x \geq 3 \\ -(x-3) & \text{if } x < 3 \end{cases}$$

So

$$\begin{aligned} \lim_{x \rightarrow 3^-} (2x + |x-3|) \\ = \lim_{x \rightarrow 3^-} (2x - x + 3) = 6 \end{aligned}$$

and

$$\begin{aligned} \lim_{x \rightarrow 3^+} (2x + |x-3|) &= 6 \\ \Rightarrow \lim_{x \rightarrow 3} (2x + |x-3|) &= 6 \text{ — left and right limits are equal.} \end{aligned}$$

$$44. \lim_{x \rightarrow -2} \frac{2-|x|}{2+x}$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \text{ So}$$

$$\lim_{x \rightarrow -2} \frac{2+x}{2+x} = 1$$

$$45. |x| = -x \text{ if } x < 0 \text{ so}$$

$$\begin{aligned} \lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right) &= \lim_{x \rightarrow 0^-} \frac{1}{x} + \frac{1}{x} \\ &= \frac{2}{x} = -\infty \end{aligned}$$

$$46. |x| = x \text{ if } x > 0 \text{ so}$$

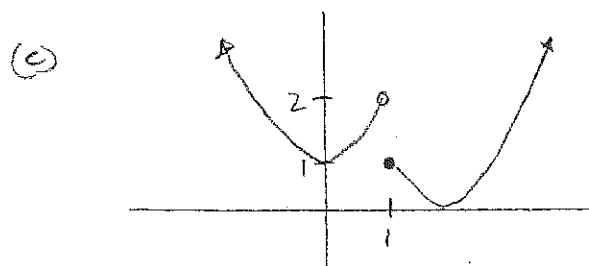
$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{|x|} \right) = \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x} \right) = 0$$

$$50. f(x) = \begin{cases} x^2 + 1 & \text{if } x < 1 \\ (x-2)^2 & \text{if } x \geq 1 \end{cases}$$

$$(a) \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2 + 1) = 2$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x-2)^2 = 1$$

$$(b) \lim_{x \rightarrow 1} f(x) \text{ DNE}$$



$$62. f(x) = \begin{cases} -1 & \text{if } x \geq 0 \\ 1 & \text{if } x < 0 \end{cases}$$

$$\therefore \lim_{x \rightarrow 0} f(x) \text{ DNE}$$

$$g(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

$$\therefore \lim_{x \rightarrow 0} g(x) \text{ DNE. But}$$

$$g(x) + f(x) = 0 \text{ for all } x$$

$$\Rightarrow \lim_{x \rightarrow 0} [g(x) + f(x)] = 0$$

Ex 3.3 p 196

$$39. \lim_{x \rightarrow 0} \frac{\sin 5x}{3x} = \lim_{x \rightarrow 0} \frac{5}{3} \left[\frac{\sin 5x}{5x} \right]$$

$$= 5/3$$

$$40. \lim_{x \rightarrow 0} \frac{\sin x}{\sin \pi x} = \lim_{x \rightarrow 0} \frac{1 \sin x}{\pi x} \cdot \frac{\pi x}{\sin \pi x} = 1/\pi$$

$$43. \lim_{x \rightarrow 0} \frac{\sin 3x}{5x^3 - 4x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{3x}{x(5x^2 - 4)}$$

$$= 1 \cdot \left(\frac{3}{-4} \right) = -3/4$$

$$44. \lim_{x \rightarrow 0} \frac{\sin 3x \sin 5x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{\sin 5x}{5x} \cdot x^{3.5}$$

$$= 15$$

Unit 2.4

Ex 2.5 p 124

3. (a) At -4 [$f(-4)$ not defined]

At -2 , 2 and 4 [limits do not exist]

(b) $\lim_{x \rightarrow -2^-} f(x) = f(-2)$ so

f continuous from the left at -2 .

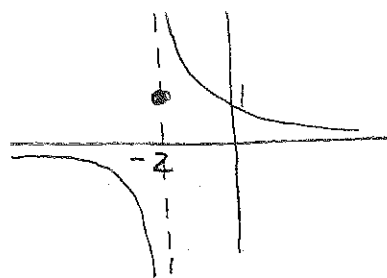
$\lim_{x \rightarrow 2^+} f(x) = f(2)$ and

$\lim_{x \rightarrow 4^+} f(x) = f(4)$ so f

continuous from the right at $x=2$ and $x=4$.

f is not continuous from either side at $x=-4$ since $f(-4)$ is undefined.

$$18. f(x) = \begin{cases} \frac{1}{x+2} & \text{if } x \neq -2 \\ 1 & \text{if } x = -2 \end{cases}$$



$f(-2) = 1$ but

$\lim_{x \rightarrow -2^-} f(x) = -\infty$ and $\lim_{x \rightarrow -2^+} f(x) = \infty$

so $\lim_{x \rightarrow -2} f(x) \text{ DNE}$ and f not continuous at $x = -2$.

$$23. f(x) = \frac{x^2 - x - 2}{x - 2} = \frac{(x+1)(x-2)}{x-2}$$

$$= x+1 \text{ if } x \neq 2$$

$\therefore \lim_{x \rightarrow 2} f(x) = 3$. Define $f(2) = 3$

then f continuous at $x=2$.

$$46. f(x) = \begin{cases} \frac{x^2-4}{x-2} & \text{if } x < 2 \\ 9x^2 - bx + 3 & \text{if } 2 \leq x < 3 \\ 2x - a + b & \text{if } x \geq 3 \end{cases}$$

$$\text{At } x=2 \quad \lim_{x \rightarrow 2^-} \frac{x^2-4}{x-2} = \frac{(x+2)(x-2)}{x-2} = 4$$

$$\lim_{x \rightarrow 2^+} (9x^2 - bx + 3) = 4a - 2b + 3$$

$$f(2) = 4a - 2b + 3 \quad \text{so}$$

f continuous at 2 if

$$4a - 2b + 3 = 4 \Rightarrow 4a - 2b = 1 \quad (1)$$

$$\text{At } x=3 \quad \lim_{x \rightarrow 3^-} (9x^2 - bx + 3) = 9a - 3b + 3$$

$$\lim_{x \rightarrow 3^+} (2x - a + b) = 6 - a + b$$

$$f(3) = 6 - a + b \quad \text{so}$$

f continuous at 3 if

$$9a - 3b + 3 = 6 - a + b \Rightarrow$$

$$10a - 4b = 3 \quad (2)$$

Solve (1) and (2) simultaneously

$$(1) \times -2$$

$$-8a + 4b = -2 \quad (3)$$

$$(2) + (3)$$

$$2a = 1$$

$$\Rightarrow a = \frac{1}{2}$$

$$\Rightarrow b = \frac{1}{2}$$

So f continuous on $(-\infty, \infty)$

if $a = \frac{1}{2}$ and $b = \frac{1}{2}$.

$$47. \lim_{x \rightarrow 2} [3f(x) + f(x)g(x)]$$

$$= 3 \lim_{x \rightarrow 2} f(x) + \lim_{x \rightarrow 2} f(x) \cdot \lim_{x \rightarrow 2} g(x)$$

$$= 3f(2) + f(2) \cdot g(2) = 36$$

$$\Rightarrow f(2)[3 + 6] = 36$$

$$\Rightarrow f(2) = 4$$

$$71. f(x) = \begin{cases} x^4 \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

f is continuous on $(-\infty, 0) \cup (0, \infty)$

since it is a product of a polynomial and a trigonometric function. Since

$$-1 \leq \sin(\frac{1}{x}) \leq 1$$

$$\Rightarrow -x^4 \leq x^4 \sin(\frac{1}{x}) \leq x^4 \quad \text{and}$$

$$\lim_{x \rightarrow 0} -x^4 = \lim_{x \rightarrow 0} x^4 = 0 \quad \text{then}$$

$$\lim_{x \rightarrow 0} x^4 \sin(\frac{1}{x}) = 0$$

by the squeeze theorem.

$$\text{and } f(0) = 0$$

$\Rightarrow f$ continuous at $x=0$

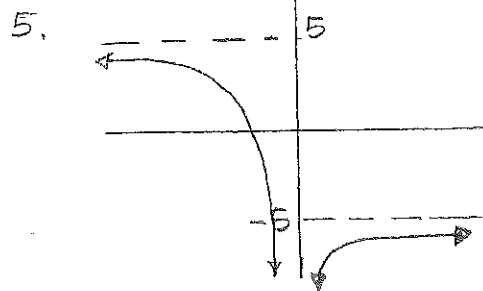
and hence on $(-\infty, \infty)$.

Unit 2.5

Ex 2.6 p137

3. (a) -2 (b) 2 (c) ∞

(d) $-\infty$ (e) Vertical $x=1$ and $x=3$; Horizontal $y=-2$ and $y=2$



7

$$\lim_{x \rightarrow -\infty} \frac{1-x-x^2}{2x^2-7} = \lim_{x \rightarrow -\infty} \frac{\frac{1}{x^2}-\frac{1}{x}-1}{2-\frac{7}{x^2}} = -\frac{1}{2}$$

19. $\lim_{t \rightarrow \infty} \frac{\sqrt{t}+t^2}{2t-t^2} = \lim_{t \rightarrow \infty} \frac{\frac{1}{\sqrt{t}}+1}{\frac{2}{t}-1} = -1$

24. $\lim_{x \rightarrow -\infty} \frac{\sqrt{1+4x^6}}{2-x^3} = \lim_{x \rightarrow -\infty} \frac{-x^3 \sqrt{\frac{1}{x^3}+4}}{x^3(\frac{2}{x^3}-1)} = 2$
 ($\sqrt{x^6} = -x^3$ for $x < 0$)

27. $\lim_{x \rightarrow \infty} (\sqrt{9x^2+x} - 3x) \times \frac{\sqrt{9x^2+x} + 3x}{\sqrt{9x^2+x} + 3x}$
 $= \lim_{x \rightarrow \infty} \frac{9x^2+x-9x^2}{x(\sqrt{9+\frac{1}{x}}+3)} = \frac{1}{6}$
 [$\sqrt{x^2} = x$ for $x > 0$]

49. $y = \frac{2x^2+x-1}{x^2+x-2}$

$\lim_{x \rightarrow \pm\infty} \frac{x^2(2+\frac{1}{x}-\frac{1}{x^2})}{x^2(1+\frac{1}{x}-\frac{2}{x^2})} = 2$ so

$y=2$ is a horizontal asymptote.
 $f(x) = \frac{2x^2+x-1}{(x-1)(x+2)}$ so that

$x=1$ and $x=-2$ are possible asymptotes.

$\lim_{x \rightarrow -2^-} \frac{(2x-1)(x+1)}{(x+2)(x-1)} = \infty$

$\lim_{x \rightarrow -2^+} \frac{(2x-1)(x+1)}{(x+2)(x-1)} = -\infty$

$\lim_{x \rightarrow 1^-} \frac{(2x-1)(x+1)}{(x+2)(x-1)} = -\infty$

$\lim_{x \rightarrow 1^+} \frac{(2x-1)(x+1)}{(x+2)(x-1)} = \infty$

So $x=-2$ and $x=1$ are vertical asymptotes.

52. $y = \frac{2e^x}{e^x-5} = f(x)$

$\lim_{x \rightarrow \infty} \frac{e^x 2}{e^x(1-\frac{5}{e^x})} = 2$ so

$y=2$ is a horizontal asymptote.
 f is not defined if

$e^x - 5 = 0$

$\Rightarrow x = \ln 5$ Consider:

$\lim_{x \rightarrow (\ln 5)^+} \frac{2e^x}{e^x-5} = \infty$ or

$\lim_{x \rightarrow (\ln 5)^-} \frac{2e^x}{e^x-5} = -\infty$

Therefore

$x = \ln 5$ is a vertical asymptote

Unit 2.6

Ex 2.7 p 148

3.(a) (i) $f(x) = 4x - x^2$ and $P(1, 3)$

$$\begin{aligned}
 m &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\
 &= \lim_{x \rightarrow 1} \frac{4x - x^2 - f(1)}{x - 1} \\
 &= \lim_{x \rightarrow 1} \frac{4x - x^2 - 3}{x - 1} \\
 &= \lim_{x \rightarrow 1} \frac{-(x-1)(x-3)}{x-1} = 2
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \\
 m &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4(1+h) - (1+h)^2 - 3}{h} \\
 &= \lim_{h \rightarrow 0} \frac{-h^2 + 2h}{h} = 2
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad y - y_1 &= m(x - x_1) \\
 y - 3 &= 2(x - 1) \\
 y &= 2x + 1
 \end{aligned}$$

12. (a) Runner A runs the entire race at a constant velocity.

Runner B starts the race at a slower velocity but finishes the race faster.

(b) Greatest distance at the time when the largest vertical line segment fits between the two graphs. Appears to be between 9 and 10 seconds.

(c) When the slopes of the two graphs are equal. This appears to be at 9.5 sec.

13. $s(t) = 10t - 4.9t^2$

Velocity when $t = 2$

$$\begin{aligned}
 v(2) &= \lim_{h \rightarrow 0} \frac{s(2+h) - s(2)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{-4.9h^2 - 9.6h}{h} \\
 &= -9.6 \text{ m/s}
 \end{aligned}$$

Instantaneous velocity when $t = 2$ is -9.6 m/s .

18. (a) On $[20, 60]$

$$\frac{f(60) - f(20)}{60 - 20} = \frac{700 - 300}{40} = 10$$

(b) Any interval that has the same y-values - $[0, 57]$ $f(0) = 600$ $f(57) = 600$

$$\text{(c) On } [40, 60] \quad \frac{f(60) - f(40)}{60 - 40} = 25$$

$$\text{On } [40, 70] \quad \frac{f(70) - f(40)}{70 - 40} = 23\frac{1}{2}$$

Avg rate of change on $[40, 60]$ is larger.

$$\text{(d)} \quad \frac{f(40) - f(10)}{40 - 10} = \frac{200 - 400}{30} = -6\frac{2}{3}$$

Value represents the slope of the line segment from $(10, f(10))$ to $(40, f(40))$

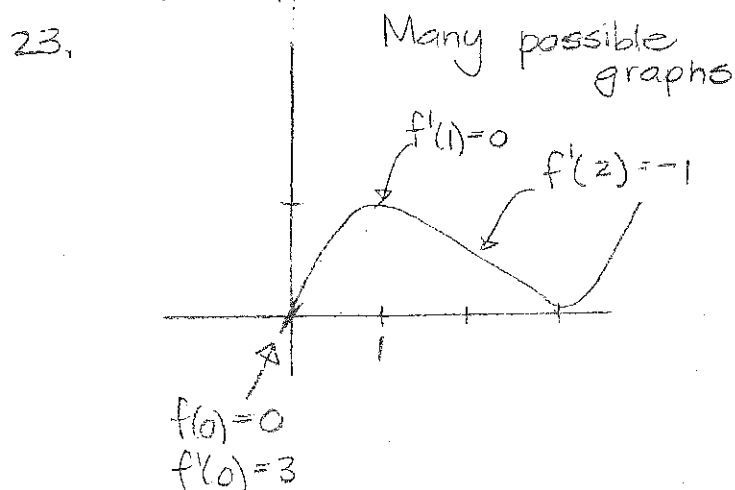
(6)

20. $g(5) = -3 \Rightarrow (5, -3)$ lies on the graph of g and $g'(5) = 4 \Rightarrow$ slope at $x=5$ is 4.

$$y - (-3) = 4(x - 5)$$

$$y = 4x - 23$$

22. $(4, 3)$ on $f(x) = y \Rightarrow f(4) = 3$. The tangent line passes through $(4, 3)$ and $(0, 2)$ so $m = \frac{1}{4}$ so $f'(4) = \frac{1}{4}$.



59. $f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(\sin \frac{1}{h}) - 0}{h}$$

$$= \lim_{h \rightarrow 0} \sin(\frac{1}{h}) \quad \text{and this}$$

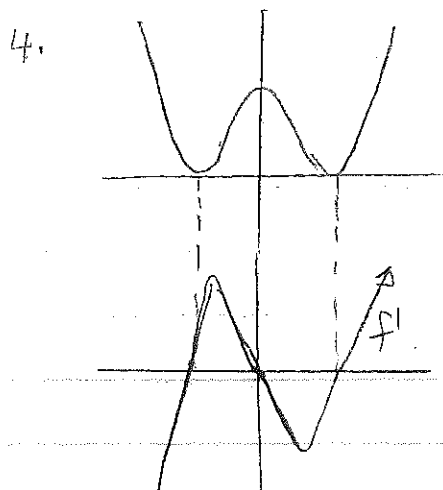
limit does not exist.

$\sin(\frac{1}{h})$ takes on values -1 and 1
see p 86 example 4

Unit 2.7

Ex 2.8 p160

3. (a) II (b) IV
(c) I (d) III



26. $f(x) = x + \sqrt{x}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h) + \sqrt{x+h} - (x + \sqrt{x})}{h}$$

$$= \lim_{h \rightarrow 0} 1 + \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$= 1 + \lim_{h \rightarrow 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= 1 + \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= 1 + \frac{1}{2\sqrt{x}}$$

Domain of $f = [0, \infty)$,
domain of $f' = (0, \infty)$

41. f not differentiable at $x = -4$ — graph has a corner. Also not at $x = 0$ since f is discontinuous at 0.

42. Not differentiable at $x = -1$ — discontinuity.
 $x = 2$ — graph has a corner